

Activity 6: Combining infinite sequences – part 2

CHALLENGE

Are all the rational numbers countable?

Task A: Producing all the positive rational numbers

*In Activity 4 you produced a sequence of the rational numbers between 0 and 1. In Activity 5 you produced a sequence of rational numbers greater than 1. Combine them using the *Merge* robot of Activity 2.*

If you ran all the robots forever would the number 1 be in the merged sequence?

Explain...

If it is, would it be in the sequence more than once?

Explain...

Task B: Producing all the *rational numbers*

Use the *Merge* robot and a robot like the *Negator* robot from Activity 2 to produce all the *rational numbers* .

Can the rational numbers be counted?

If not, why not?

If so, how would you convince someone?